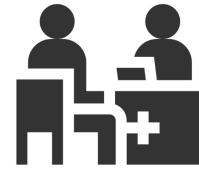
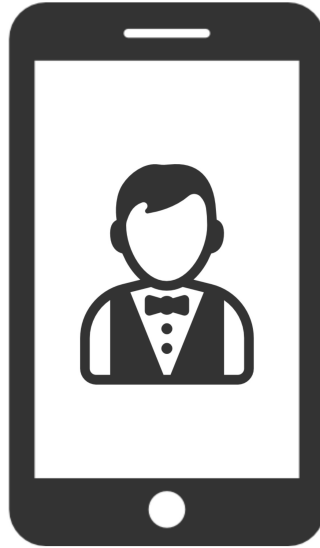


Integrating our data in a decentralized Web ecosystem

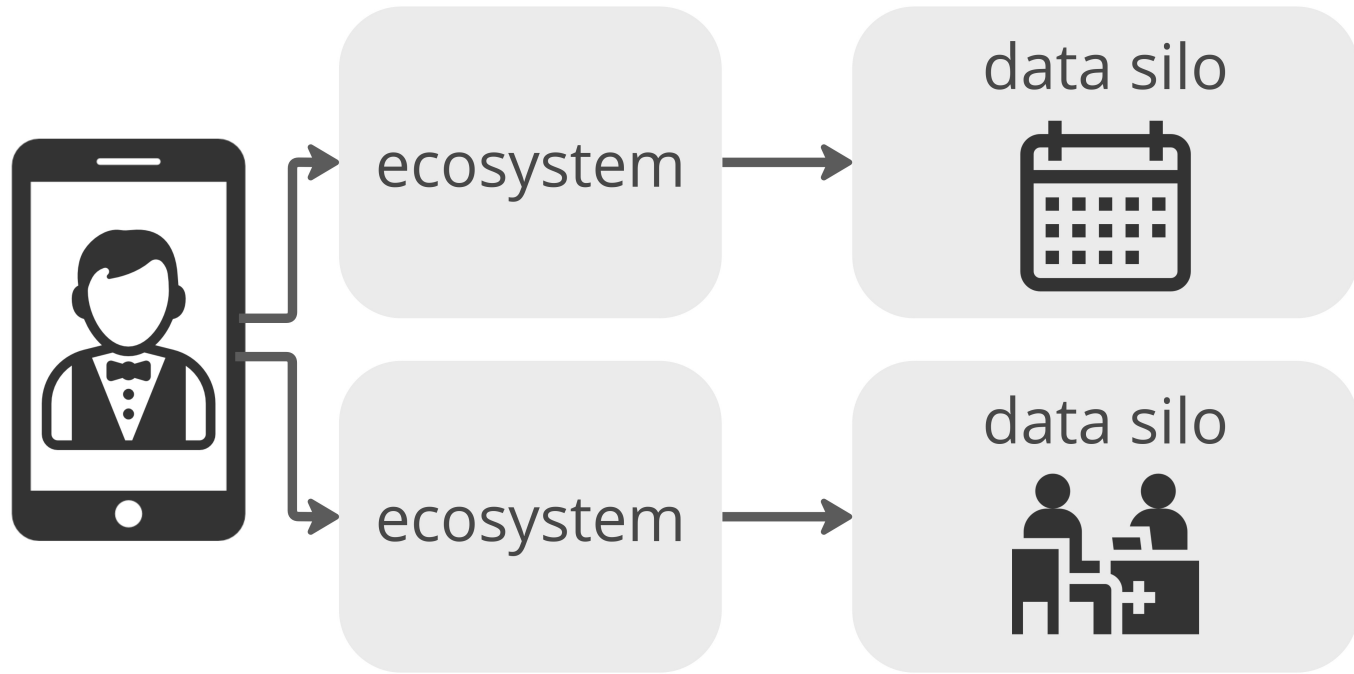
Ruben Dedecker



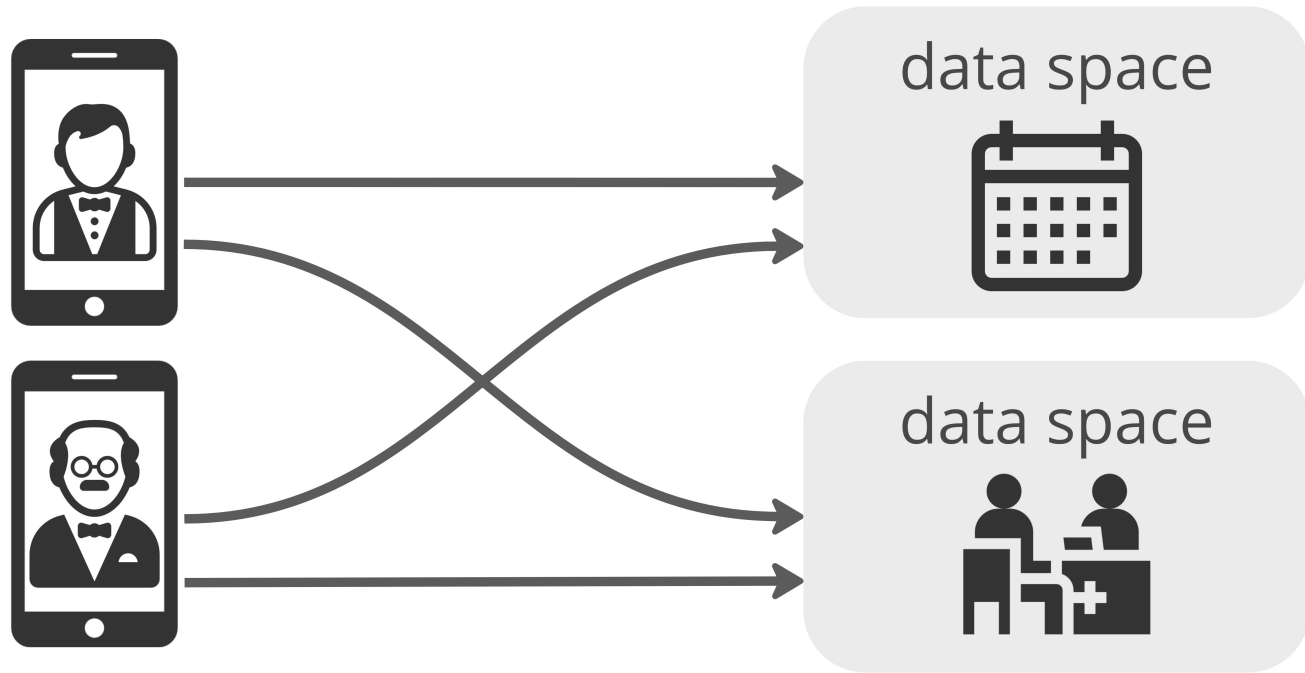
Where is our butler?



The data our butler needs is
stuck in closed ecosystems



We can publish our data on decentralized data spaces



I am part of the KNoWS lab

Our lab works on
knowledge on Web scale

My research focus is on the
publishing of personal data

**Integration of open data
is an optimization problem**

**Integration of personal data
is a governance problem**

We need to extend our data spaces with governance functionality

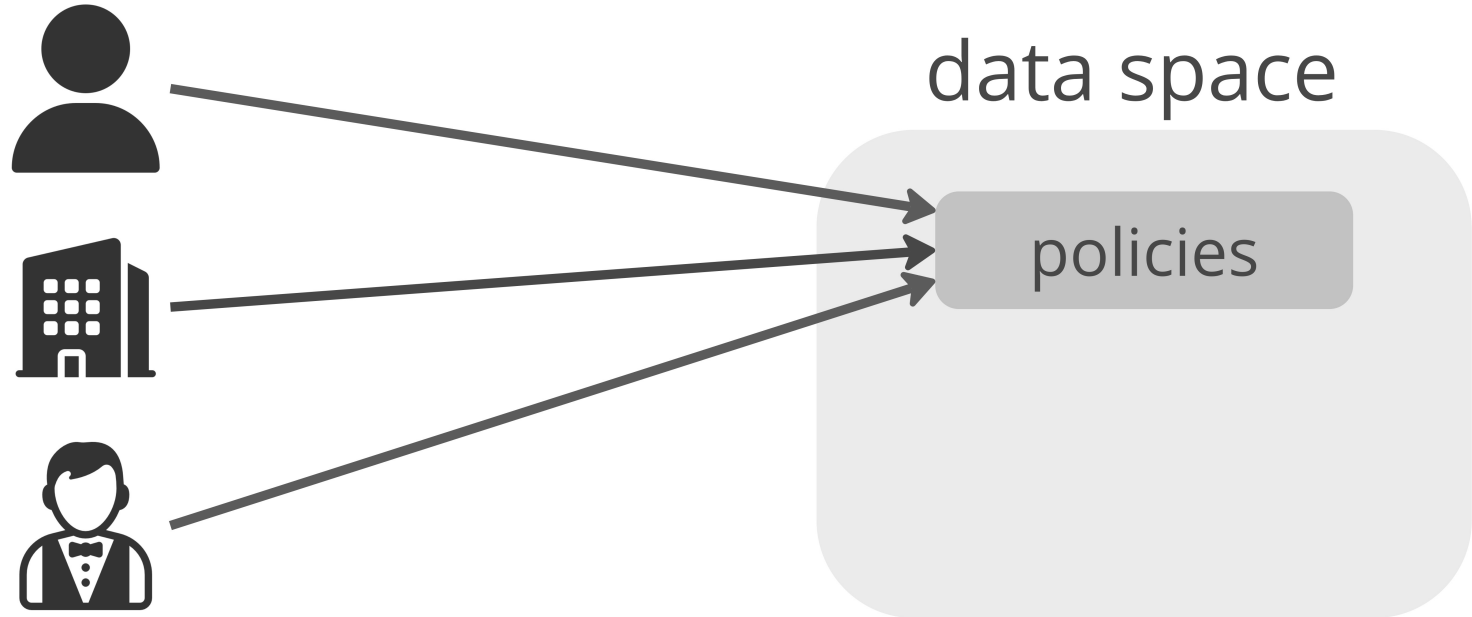
We require **control** of our data

We require **usability** of our data

We need to **track usage** of our data

We need control over our data

Policies enforce rules and preferences



Policies provide flexibility in data selection

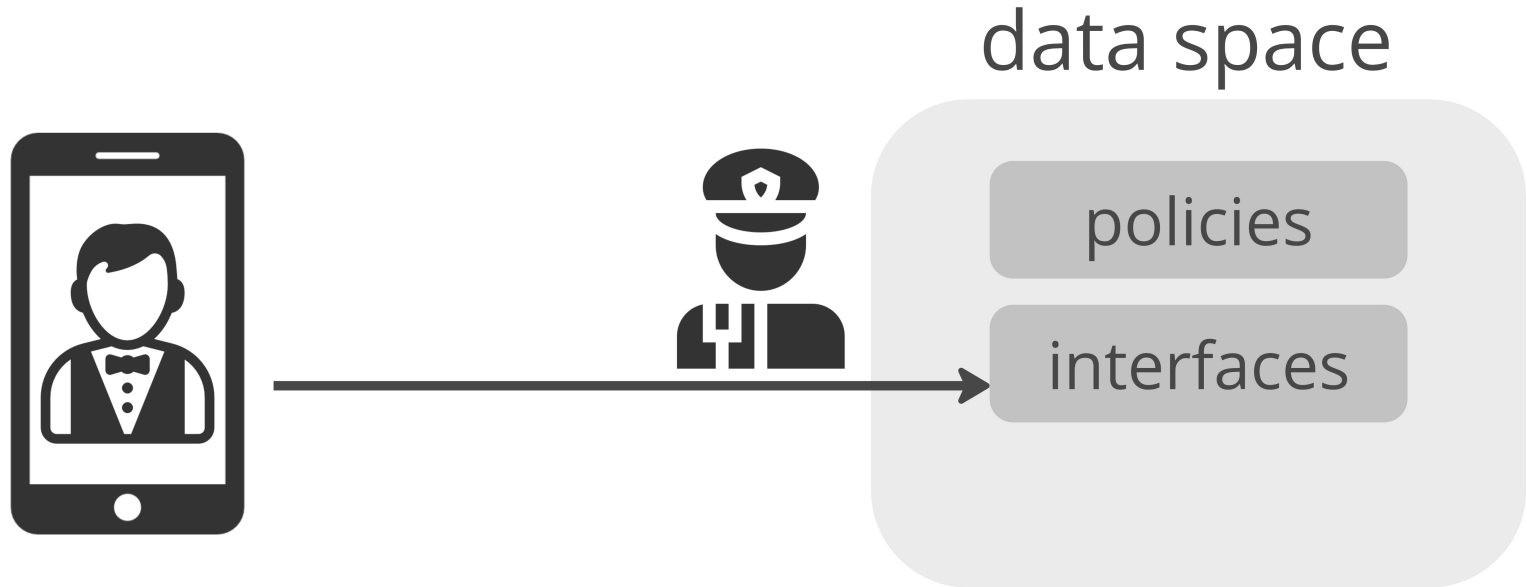
Policies enable static and dynamic conditions

Agents can negotiate for exceptions

Reasoning provides proof

We need our data to be usable

Interfaces reconcile access and control



Interfaces must reflect our policy requirements

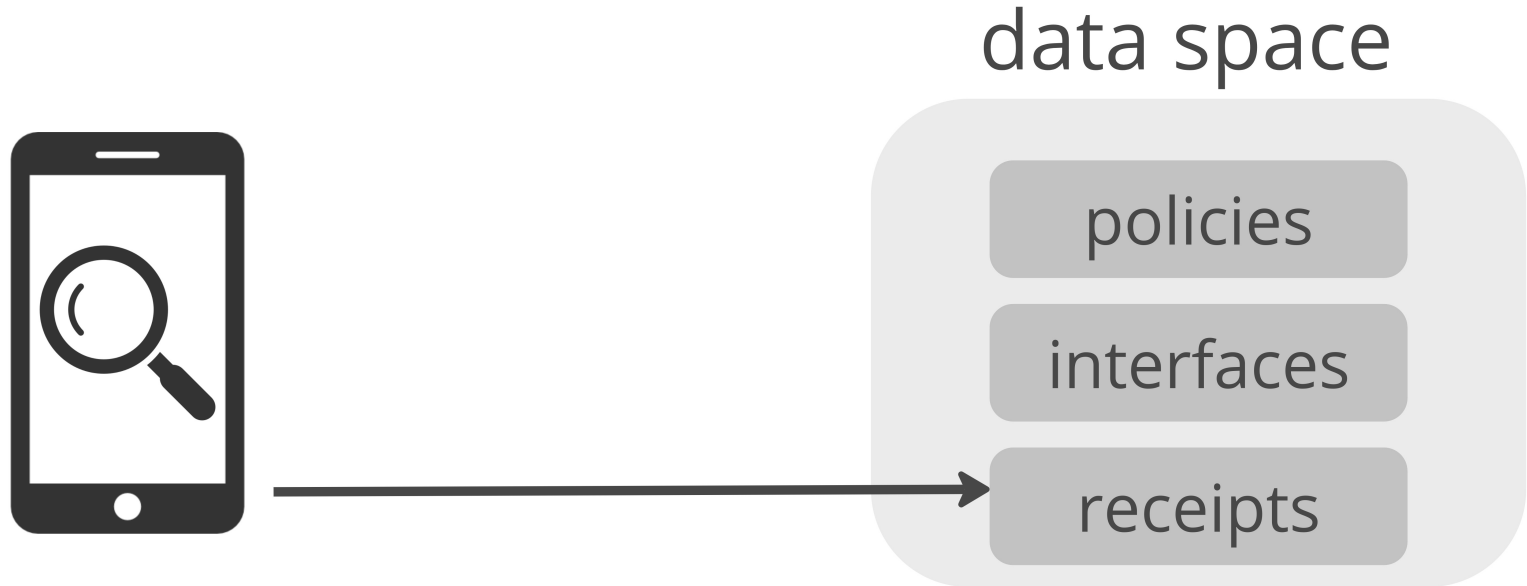
Interfaces provide only requested data

Interfaces enable discovery of additional data

Reasoning provides proof

We need to track usage of our data

We store receipts of data exchanges



Receipts give an overview of data exchanges

Receipts allow us to detect misuse

We create a feedback loop to our policies

Receipts provide provenance

Signatures provide trust

We can automate integration of our personal data

control

Policies help us trust
agents in our data space

usability

Interfaces help agents find
relevant data in our space

usage

Receipts lets us track data usage
and adapt our policies accordingly

There is our butler!

